

Inception

Introduction

Inception is the beginning phase of requirements engineering. It defines how does a software project get started and what is the scope and nature of the problem to be solved. The goal of the inception phase is to identify concurrence needs and conflict requirements among the stakeholders of a software project. To establish the groundwork we have worked with the following factors related to the inception phases:

1. Identifying Stakeholders
2. Recognizing multiple viewpoints
3. Working towards collaboration
4. Asking the First Questions

Identifying Stakeholders

Stakeholder refers to any person or group who will be affected by the system directly or indirectly. Stakeholders include end-users who interact with the system and everyone else in an organization that may be affected by its installation. To identify the stakeholders we consulted with Assistant Librarian(Program) and asked her following questions:

- Who is paying for the project?
- Who will be using the project outcomes?

- Who gets to make the decisions about the project (if this is different from the money source)?
- Who has resources I need to get the project done?
- Whose work will my project affect? (During the project and also once the project is completed).

Concluding thoughts on Stakeholders, We identified following stakeholders for our automated book circulation system of a library:

1. **The University Librarian (Project Sponsor):** The University Librarian is the person who has the final authority over our budget, our personnel resources, and ultimately the finished product. His position empowers him to veto a decision made by the other Stakeholders.
2. **The Associate University Librarian (Systems Head):** As head of Library Systems, the Associate University Librarian has direct authority over our Systems budget, and our team — the people developing the site and doing much of the “management” end of this project.
3. **The Circulation Manager(Adminstrator):** The Circulation Manager has the adminstrative power to handle the software. So he is also our stakeholder.
4. **System Operator:** System Operator will directly interact with this software. So they are also a major stakeholder of our system.
5. **Borrowers:** We will provide some features where bborrowers will directly interact with this software.
6. **Developers:** We selected developers as stakeholder because they develop this system and work for further development. If occurs any system interruption, they will find the problem and try to solve it.
7. **University:** University will finance the project and it has some has rules and regulation to maintain our system. We have to follow them strictly. So university is

also our stakeholder.

Recognizing multiple viewpoints

1. The University Librarian (Project Sponsor) view points:

- User friendly and efficient system
- Minimum maintenance cost
- Availability of expected requirements within the budget

2. The Associate University Librarian (Systems Head) view points:

- Allow the system to be accessed via the Internet.
- Restrict access to functionality of the system based upon user roles
- The application can be accessed from any computer that has Internet access
- Web-Based Interfaces

3. The Circulation Manager(Administrator) view points :

- Allow Librarians to check-out and check-in items for valid users.
- The application only needs to be installed and maintained on one computer.
- Strong Authentication system
- Allow Librarians to generate reports on the items in the system (e.g., all overdue items, all missing items.)

4. System Operator view points :

- Easy to operate
- User friendly
- Automatic generate reports on the items in the system
- A user guide describing how to use LCS need to be deployed with the system.
- A product reference manual describing how to install, setup, and run the application shall be provided.

5. Borrowers view points:

- Allow the system to be accessed via the Internet.
- User friendly and efficient systems
- Easy Access
- Allow any user to search for items
- Allows valid users to renew items online by logging into the system

6. Developers viewpoints:

- Easy to develop
- No ambiguous requirement

7. University viewpoints:

- Buy the system within their budget
- No disruption of rules and regulation

Working towards collaboration

Every stakeholder has their own requirement. In this step, we merged these requirements. We followed following steps to complete the task:

- Identify the common and conflicting requirements
- Categorize the requirements
- Take priority points for each requirements from stakeholders and on the basis of this voting prioritize the requirements
- Make final decision about the requirements.

Common requirements:

- User friendly and efficient system
- Easy to operate
- The application can be accessed from any computer that has Internet access

Conflicting Requirements:

- Minimum maintenance cost
- Availability of all requirements within the budget
- Easy access
- Restrict access to functionality of the system based upon user roles
- No ambiguous requirement

We finalized following requirements for the system by categorizing and prioritizing the requirements.

Final Requirements:

1. Error free system (Maximum 5% error may be considerable)
2. Accessible via the Internet.
3. Allow valid users to login and logout.
4. Restrict access to functionality of the system based upon user roles
5. Allow administrators of the system to change user types and configure parameters of the system
6. Allow any user to search for items in the library's circulation without having to log in to the system
7. Allow valid users that log in to renew items, reserve items, and view the items they have checked out.
8. Allow Librarians to check-out and check-in items for valid users

9. Allow Librarians to generate reports on the items in the system (e.g., all overdue items, all missing items.)
10. The application only needs to be installed and maintained on one computer
11. Allows valid users to renew items online by logging into the system

Asking the First Questions

We set our first set of context-free questions focuses on the customer and other stakeholders, overall project goals and benefits. The questions are mentioned above. These questions helped us to identify all stakeholders, measurable benefit of the successful implementation and possible alternatives to custom software development. Next set of question helped us to gain a better understanding of problem and allows the customer to voice his or her perception about the solution. The final set of question focused on the effectiveness of the communication activity itself.

Conclusion

Inception phase helped us to establish basic understanding about book circulation system in a library, identify the people who will be benefited if book circulation system becomes automated, define the nature of the book circulation software and establish a preliminary communication with our stakeholders.

